

KEVIN OMONDI OBOTE

AI Researcher & Systems Engineer | Technical Product Manager | Data Scientist | Teaching & Mentorship

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PROFESSIONAL PROFILE

AI Systems Researcher, Data Scientist, Technical Product Manager, and Educator with over 5 years of experience spanning academic research, industry AI systems, SaaS product management, and technical mentorship. My research focuses on how intelligent systems can generalise beyond training distributions through online adaptation, neuro-symbolic integration, and learning under data scarcity, with published peer-reviewed work in low-resource NLP, machine learning, and spatio-temporal analytics. I have developed production-ready AI pipelines for fintech, inclusive AI, and multilingual African language systems, and led the full SaaS product lifecycle for ERP platforms deployed across East and West Africa. My work includes establishing the first documented ASR baseline for Kalenjin (accepted, IEEE Xplore AAIAC 2026), building award-winning inclusive AI systems, and facilitating technical education for learners across Africa. Equally comfortable contributing to industry product teams, academic research environments, and educational institutions.

EDUCATION

MSc, Data Science and Analytics (Computational Statistics)

May 2024 – May 2026

Strathmore University – Nairobi, Kenya

- Relevant coursework: Statistical Learning Theory, Bayesian Statistics, Probability & Stochastic Processes, Time Series Analysis, Advanced Algorithms, AI Ethics, Optimisation for Data Science
- Research focus: spatio-temporal learning, adaptive modelling, responsible AI governance

Data Science Certificate

May 2023 – July 2024

ALX – Explore Data Science Academy – Nairobi, Kenya

- Intensive programme covering machine learning engineering, MLOps, and deep learning applications

BSc, Mathematics and Computer Science (Pure Mathematics)

September 2017 – April 2023

Machakos University – Machakos, Kenya

- Foundations: real analysis, abstract algebra, topology, functional analysis, linear algebra, probability theory, numerical methods, differential equations, mathematical statistics

CORE COMPETENCIES

- AI & Machine Learning:** Probabilistic Modelling, Diffusion Models, GNNs, LSTMs, Transformers, LLM Post-Training, Meta-Learning (MAML), Online Learning, Reinforcement Learning, Agentic AI
- Mathematical Foundations:** Probability Theory, Bayesian Inference, Stochastic Processes, Functional Analysis, Linear Algebra, Abstract Algebra, Topology, Statistical Inference
- Applied AI & Fintech:** Credit Scoring & Alternative Data Modelling, Financial Inclusion Analytics, Decision Support & Risk Modelling, Real-Time AI Pipelines
- Responsible AI:** Explainability (SHAP, LIME), Fairness & Bias Mitigation, AI Governance Frameworks, Data Privacy, Ethical Compliance
- ML Engineering & MLOps:** FastAPI, Python, PyTorch, HuggingFace Transformers, GCP, Azure, Data Pipelines, Model Deployment & Monitoring, Versioning
- Teaching & Mentorship:** Curriculum Design, Technical Training, 1:1 Mentoring, Group Facilitation, CRISP-DM, Data Analysis, Machine Learning
- Product Management:** Agile/Scrum, Roadmap Development, Stakeholder Management, Cross-functional Leadership, PRDs, User Stories

TEACHING & COMMUNITY LEADERSHIP

Course Facilitator & Teaching Assistant

May 2024 – May 2026

iLabAfrica, Strathmore University – Nairobi, Kenya

- Served as a teaching assistant, facilitator, and course supporter for the CDAV, ICDS, and Professional AI programmes, supporting over 100+ working professionals

- Facilitated instruction in data analysis, data science, machine learning applications, history of AI, CRISP-DM methodology, and foundational statistics
- Delivered practical lessons on transformer architectures, data simulation, and applied ML, contributing to peer-validated journal publication outputs

Data Analysis Path Lead (Volunteer)

September 2024 – September 2025

Genesys Tech Hub – Pan-African (Remote)

- Designed and developed the data analysis curriculum and course content for learners across Africa
- Mentored and tutored data analysis students, providing personalised 1:1 guidance and technical support
- Coordinated with fellow mentors and tutors to ensure cohesive and effective content delivery across the programme

RESEARCH EXPERIENCE

Research Scholar

May 2024 – May 2026

iLabAfrica, Strathmore University – Nairobi, Kenya

- Leading autonomous modelling research for large-scale analytics, with focus on adaptive time-series and spatio-temporal systems
- Conducting Responsible AI and AI governance research, ensuring ethical compliance of deployed AI systems in sensitive domains (finance, public policy)
- Developing adaptive modelling frameworks for learning under distributional shift and data scarcity
- Collaborating with multidisciplinary teams across mathematics, computer science, and biostatistics to develop and validate statistical methods for real-world data

Applied Research – Alternative Credit Scoring & Financial Inclusion

August 2025 – November 2025

iLabAfrica, Strathmore University – Nairobi, Kenya

- Applied research on credit scoring for unbanked populations using national financial access datasets (FinAccess)
- Modelled non-traditional financial signals: mobile money usage, digital access, informal income behaviour
- Implemented SHAP and LIME explainability guardrails to eliminate proxy discrimination and ensure ethical compliance
- Presented findings at fintech and AI innovation forums, emphasising fairness, bias mitigation, and responsible AI deployment

Research Enumerator

January 2025 – February 2025

Nutrition International – Kiambu, Kenya

- Supported endline surveys in Kiambu County; reviewed data instruments and improved survey quality
- Trained community health workers, strengthening data reliability in real-world deployments

PUBLICATIONS & RESEARCH

Peer-Reviewed Journal Articles

Obote, K., Kikwai, B., Senagi, K., Njiiri, J., Olukuru, J., & Sevilla, J. (2025).

- Bridging Swahili Communication Gaps: Real-Time Audio-to-Text Sentiment Analysis via Pre-trained NLP.
- American Journal of Artificial Intelligence, 9(2), 167–185. DOI: 10.11648/j.ajai.20250902.18
- Designed an end-to-end pipeline leveraging Wav2Vec2 and T5 models; achieved WER of 0.19 in low-resource Swahili speech recognition

Makokha A., Obote, K., Muchiri H., Senagi K. (2026).

- Predicting Customer Churn in the Telecommunications Industry using Machine Learning Techniques.
- American Journal of Networks and Communications, Vol. 15(1), pp. 10–26. DOI: 10.11648/j.ajnc.20261501.12
- Built a full ML pipeline on 8,454 real B2B telecom accounts; benchmarked 10 classifier families; deployed EasyEnsembleClassifier with SVM SMOTE resampling; post-hoc explainability via SHAP and LIME

Obote, K., Kikwai, B. (2026).

- The Case for Swahili NLP: Addressing Linguistic Gaps in a Vital African Language.
- Arusha Working Papers in African Linguistics
- Critical synthesis of data scarcity, tokenisation challenges, and evaluation gaps in Swahili NLP; advocates for open-source tools and cross-sector collaboration

Conference Papers (Accepted)

Obote, K., et al. (2026).

- Low-Resource ASR for Kalenjin: Two-Stage Fine-Tuning of Wav2Vec2-XLS-R with KenLM Integration.
- Accepted – AAIAC 2026: 2nd International Conference on the Advancements of Artificial Intelligence in the African Context, Dar es Salaam, Tanzania, July 22–23, 2026. To be published in IEEE Xplore.
- Developed a two-stage fine-tuning strategy on Wav2Vec2-XLS-R for Kalenjin, an under-resourced African language; integrated KenLM achieving a 55.3% relative CER reduction and 10.7% WER improvement; establishes the first documented ASR baseline for Kalenjin

Under Review

Owino, S.A., Obote, K., et al. (2026).

- Enhancing Travel Time Prediction in African Public Transit Systems. Spatio-Temporal Discovery Journal. [Under Review]
- Developed a hybrid Graph Neural Network (GNN) + LSTM architecture modelling complex spatio-temporal dependence structures in urban transit networks

INDUSTRY & AI SYSTEMS EXPERIENCE

Lead AI Systems Engineer

February 2023 – Present

Guild Code – Nairobi, Kenya

- Architected and deployed production-ready agentic AI pipelines capable of autonomous decision-making and online learning under noisy, incomplete data conditions
- Sema Sasa (Inclusive AI): Designed a native ASR system addressing linguistic exclusion for low-resource East African languages(Kiswahili) – Winner, Deep Learning Indaba Community Challenge 2025
- Developed multilingual, code-switching AI agents for informal economic sectors, serving as a primary driver of regional financial inclusion
- Built eKuza: an AI-driven agribusiness management platform digitising smallholder farmer tracking and real-time payment automation
- Championed an OpenSourceFirst architectural philosophy, advising regional startups on data sovereignty and preventing vendor lock-in

Associate Product Manager

May 2023 – June 2024

eProd Solutions Limited – Nairobi, Kenya

- Owned the full SaaS product lifecycle for ERP modules used by 20+ agribusinesses across East and West Africa, managing roadmap, prioritisation (MoSCoW/RICE), and cross-functional delivery
- Integrated data-driven forecasting and predictive logistics systems into ERP platforms; utilised Power BI and SQL to analyse user data, driving a 20% improvement in system accuracy and user retention
- Translated complex logistics and payment workflows into technical user stories for engineering teams; coordinated country-specific feature updates for compliance with regional regulatory environments
- Unified fragmented data modules (inventory, logistics, sales) into single-source-of-truth architectures, eliminating 40% of manual intervention overhead and cutting reporting burdens significantly
- Initiated a Data Literacy programme for interns and clients, reducing onboarding time by 30%; directly mentored 3 product interns into full-time technical and data roles

Election Project Manager (Data & System Operations)

February 2022 – March 2023

Roman Solutions Limited – Machakos, Kenya

- Managed digital infrastructure for 7,000+ concurrent users; ensured system integrity, monitoring, and rapid incident response under mission-critical, compliance-heavy conditions
- Led cross-functional team under high-pressure timelines, achieving 100% execution of product deliverables
- Conducted training and outreach sessions for non-technical stakeholders; enforced strict regulatory and audit requirements

EARLY CAREER EXPERIENCE

- Data Analyst & Client Officer – eProd Solutions (September 2022 – May 2023)
- Software Engineer Intern – eProd Solutions (January 2022 – February 2023)
- IT Intern – Swahilipot Hub Foundation (January 2021 – April 2021)

AWARDS & HONOURS

- **Winner** – Deep Learning Indaba Community Challenge 2025, Kigali, Rwanda (Sema Sasa: Inclusive AI System)
- **Finalist** – Data Science Africa (DSA) Hackathon 2025, Ibadan, Nigeria
- **Research Scholar Award Scholarship** – iLabAfrica, Strathmore University

REFERENCES

Dr. Evans Omondi Associate Research Scientist APHRC, Nairobi Kenya eomondi@strathmore.edu +254735743609	Dr. Betsy Muriithi Research Fellow iLabAfrica, Strathmore University bmuriithi@strathmore.edu +254750941086	Dr. Benjamin Kikwai Lecturer : Pure Mathematics Machakos University kikwaib@mksu.ac.ke +254715362549
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